

TTAM: Additional Experiments

Execution plan for the revised Section 6 | 8 September 2026

Objective. Evaluate the complete AMG-TP + AP-OPS pipeline under one chronological protocol, quantify the contribution of each module, and measure downstream bidding value. The target is a six-method main comparison, a four-variant ablation, and bidding replay using the same predictions. This document specifies work to run; it does not report new results.

1. Existing work and remaining work

Component	Available	Required supplement
Prediction evaluation	Full-data Criteo/Avazu runners, expert models, baseline implementations, AP-OPS component results.	Integrate full TTAM and rerun with all tuning restricted to past data.
Ablations	Separate AMG-TP and AP-OPS controls; 14 controlled shift regimes.	Run the four component variants on the same real streams and origins.
Bidding	AMG-TP replay, common pacing, spend/value curves and paired comparisons.	Replay full TTAM and the final baselines; add paired uncertainty.

2. Correct the selection protocol before running

The existing “nested” runner inherits ONS and meta-learning settings from development windows that overlap early outer evaluation days. Correcting only the per-origin mixture and memory settings is insufficient. All data-tuned settings must be chosen from data available before the evaluated origin. [1–3]

Dataset	Full stream	Outer test days	Inner validation at day d
Criteo Attribution	16,468,027 rows; 31 days	16–30 (15 origins)	Three latest eligible days before d
Avazu	40,428,967 rows; 10 days	5–9 (5 origins)	Three latest eligible days before d

All day indices are zero-based. The inherited tuning windows were Criteo 16–21 and Avazu 5–6. Do not reuse their selected parameter values in early origins merely because those values are stored in a frozen configuration. Retain the outer dates above and reselect the affected settings within each origin.

3. Chronological training, validation, and evaluation

For each origin d: construct candidate predictions causally over the available prefix; compute validation loss on the last three eligible days; choose one configuration shared by seeds 0, 1, and 2; replay that selected configuration over the preceding stream to initialize its state; then evaluate day d. Candidate validation predictions must be generated before observing their own labels.

A label is eligible only once its 30-minute delay has elapsed. Apply this rule to backbone training, gates, historical loss summaries, hyperparameter selection, and calibration. At midnight, omit labels that have not matured from selection and training; let their queue entries mature later. Preserve the queue across days. Use 15-minute update blocks for Criteo and one-hour blocks for Avazu.

Use mean validation log loss across the three seeds for selection; within a seed, pool eligible validation impressions. Record exact training dates, validation dates, maturity cutoffs, candidate grids, scores, choices, and replay initialization. Earlier outer days may inform later origins only after their labels become available.

Methods and shared execution

4. Freeze the implementation to match the manuscript

Use one logistic-regression backbone with L_2 regularization and the same hashed features for every comparable arm. Exclude Avazu identifiers and timestamps from predictors. Build a shared bank with nominal horizons $\{1, 3, 7, 14, \text{expanding}\}$; each predictor uses only available mature history. When a window exceeds the history length, truncate it. In Avazu, 14-day and expanding windows coincide, and other windows can coincide early. Record distinct effective horizons and handle duplicate experts consistently.

Implement AMG-TP with sample-specific gating, learned persistence, and period-level updates. Update historical memory from the weights actually deployed in the previous period, matching Section 4. Do not recompute those past weights after retraining the gate. AP-OPS receives AMG-TP predictions and combines reset, short-memory, and long-memory full Platt calibrators. Keep both slope a and intercept b trainable.

For arms without AMG-TP, fit a single constant simplex weight vector to past validation predictions and freeze it during day d . Use the same selected weights for the fixed-mixture, AP-OPS-only, and OPS arms. This control removes the AMG-TP module; ordinary backbone retraining and per-origin model selection remain common.

5. Nine unique method variants

Variant	Historical prediction	Calibration	Use
Expanding	Expanding-history expert	None	Main
Best Fixed Window	Horizon selected on inner validation; fixed during day d	None	Main
ARW	Causal adaptive window selection	None	Main
AdaMoE	Causal mixture updated from mature past losses	None	Main
OPS	Validation-fitted fixed mixture	Daily-reset OPS; free a, b	Main
TTAM	AMG-TP	AP-OPS	Main + ablation
Without both modules	Validation-fitted fixed mixture	None	Ablation
AMG-TP only	AMG-TP	None	Ablation
AP-OPS only	Same validation-fitted fixed mixture	AP-OPS	Ablation

Use identical expert predictions, evaluation impressions, origin dates, and seed IDs across comparable arms. The AMG-TP-only and full-TTAM arms share their selected historical module. The AP-OPS arms use the same algorithm and tuning rule, with settings selected on each arm's own input stream. These comparisons estimate the value of each module with past-only tuning, rather than holding a configuration optimized for one input artificially fixed on another.

6. Hyperparameter selection and checks

Freeze candidate grids, search budgets, objective, and deterministic tie-breaking before inspecting the new results. Reuse existing grids where applicable, but rerun selection at each origin for ONS regularization and step size, meta learning rate, memory/aggregation settings, OPS settings, gate/persistence settings, and baseline parameters. Any backbone setting previously tuned on later data must also be reselected or fixed independently of those data.

For affordable staged selection, first select the historical module using uncalibrated validation loss, then select calibration settings on its causal prediction stream. Use the same staged rule for all relevant arms and document it; do not claim joint optimal tuning. Cache expert predictions and share historical replay across downstream arms where configurations coincide.

Before full runs, verify causal invariance to future-label changes, cross-midnight label maturation, simplex and parameter constraints, and the $\lambda = 0$ AP-OPS-to-OPS identity. Verify the deployed-weight memory update. Reuse the existing tests where applicable and add only checks needed for the integrated pipeline.

Statistics and manuscript outputs

7. Estimands and paired uncertainty

Let $L_{m,d,s}$ be the mean impression log loss of method m on origin day d and seed s . Define seed-averaged daily loss $A_{m,d}$, the equal-day score A_m , daily gain $G_{m,d}$, and mean gain G_m by:

$$\begin{aligned} A_{m,d} &= (1/3) \sum_{s=0}^2 L_{m,d,s}, & A_m &= (1/D) \sum_d A_{m,d}. \\ G_{m,d} &= A_{m,d} - A_{\text{TTAM},d}, & G_m &= (1/D) \sum_d G_{m,d}. \end{aligned}$$

Positive gain favors TTAM. Report the mean score, mean paired gain, a 95% paired day-bootstrap interval for the gain, and origins won. Use 10,000 paired resamples and a fixed resampling seed (20260908). Apply the same sampled days to both methods, after averaging seeds within each day. Do not treat impressions or seed-by-day cells as independent temporal replicates.

Chronological days can be dependent. Report a moving-block bootstrap with block length two as a sensitivity analysis, alongside all day-level gains. With only five Avazu origins, both interval estimates are descriptive and fragile; avoid significance stars and universal superiority language. More seeds measure training variation but do not replace additional dates. If seed standard deviations are shown, label them separately from confidence intervals.

Retain impression-weighted log loss, Brier score, and calibration error as secondary appendix metrics. State the weighting and calibration-bin definitions. Do not mix an impression-weighted score with an equal-day difference without labeling the distinction. Existing dates have already informed model development: the corrected replay addresses chronological selection, not prospective independent confirmation.

8. Required main-text outputs

Section	Output	Content and discussion
6.1 Setup	Concise protocol description	Data, split including validation, mature-label rule, metrics, and methods. Refer to the frozen run manifest for reproducibility.
6.2 Results	One table + one figure	Table: six methods, mean log loss, paired gain and 95% interval, origins won. Figure: two dataset panels with origin-level gains and mean-interval overlays for each comparator.
6.3 Ablation	One four-variant table	Both modules off, AMG-TP only, AP-OPS only, and TTAM. Show scores and paired gain of TTAM over each ablation, with 95% intervals.
6.4 Bidding	One comparison table	Matched-spend clicks/value gains, paired intervals, realized spend, and replay assumptions.

Section 6.2 discussion: one paragraph stating gain sizes, consistency across origins and datasets, and exceptions. Use “robustly outperforms” only to the extent supported by the observed comparisons and uncertainty. The figure must retain unfavorable origins; it is not a selected collection of successful cases.

Section 6.3 discussion: one paragraph explaining the added value of AMG-TP given AP-OPS, and AP-OPS given AMG-TP. The primary contrasts are $A_{\text{AP-OPS only}} - A_{\text{TTAM}}$ and $A_{\text{AMG-TP only}} - A_{\text{TTAM}}$. Report whether the direction or magnitude differs across datasets. Do not infer synergy merely because the full method has the lowest mean.

9. Reuse existing component studies in the appendix

Move persistence sweeps, learned-weight trajectories, detailed shift-regime tables, and optimizer/slope controls to the appendix. Recompute synthetic confirmation summaries using the twelve confirmation seeds separately from the five development seeds. Add uncertainty from saved outcomes where possible. Results produced with recomputed historical weights must retain that implementation label; they do not automatically validate the revised deployed-weight version. New broad sweeps are not required for this plan. [4–5]

Downstream bidding and completion

10. Bidding replay using final predictions

Use the final chronological predictions from the nine-variant run; the main bidding table compares TTAM with the same five headline baselines. Freeze the auction and pacing policies across methods. Use common impression order, prices, budgets, and random draws for each paired comparison. Change only the prediction input; do not tune the bidding policy separately for TTAM.

Criteo: use the available recorded display-cost field as the price proxy required by the existing replay model. Document its meaning, cleaning, and the assumed auction rule; do not describe it as an observed clearing price unless supported by the data. **Avazu:** any generated price model is a simulation and must be labeled accordingly. Freeze its specification using past data and never use held-out click labels to generate prices. If an adequate price specification is unavailable, keep Avazu bidding pending and report Criteo replay as the primary downstream experiment.

Use a primary budget fraction of 25% of the fixed reference cost of all eligible impressions in the evaluated stream. This is an offline normalization; specify the denominator and whether budgets reset daily. For the main analysis, use daily resets and the same budget per paired origin. Report actual spend as a fraction of the reference cost and realized clicks, plus the percentage gain over each baseline.

Select bid multipliers and any pacing settings on past validation data. If actual spend differs across methods, use the existing value-versus-spend curves and a predeclared multiplier grid to interpolate to the common target. Bracket using spend only; never choose multipliers using test clicks or value. Label interpolated results explicitly, report bracketing spends, and do not extrapolate beyond the attained range. Interpolation is an evaluation normalization, not a deployed policy.

Compute uncertainty by pairing days after averaging training seeds and shared simulation draws within each day. Recompute the aggregate click-gain ratio in every paired day resample; keep simulation randomness separate from temporal variation. Show gain intervals and spend diagnostics. Report all evaluated regimes, including losses. Additional 10%, 50%, and 75% budget summaries belong in the appendix if the saved frontiers support them; no separate model training is needed. [6]

11. Execution order and completion criteria

Order	Work	Required output
1	Freeze corrected protocol and implementation	Configuration grids; maturity and split rules; algorithm revision; passing causal checks.
2	Run the shared bank and nine variants	Per-origin, per-seed metrics; selected configurations; causal state records and final predictions.
3	Compute paired comparisons	Main/ablation tables; two-panel figure; bootstrap sensitivity; secondary metrics.
4	Replay final predictions for bidding	Click gains and intervals; actual spend; price/policy definitions.
5	Fill the manuscript placeholders	One paragraph per result subsection; claims calibrated to results; appendix consolidation.

Archive code revision, dataset counts, eligible-label counts, effective horizons, seed IDs, selected settings, per-day predictions/metrics, and resampling settings. Use the same output records for tables and figures. Completion means the experiments and their limitations are reported, regardless of whether every comparison favors TTAM.

Source files for reuse and audit

- [1] Hyperparameter selection | [2] Original split definitions
- [3] Current rolling-origin runner | [4] AP-OPS findings
- [5] AMG-TP robustness report | [6] Existing bidding report